

A SCHEME FOR CONSERVATIVE INTERPOLATION ON OVERLAPPING GRIDS*

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Abstract. This paper describes how to interpolate in a conservative manner when solving systems of conservative laws on overlapping grids. Overlapping grids are a flexible and efficient way to create grids for complicated regions. Before now, however, no general method had been developed for conservative interpolation. The basic idea consists of assuming that the interpolation coefficients are free parameters and then deriving constraints to ensure that the interpolation is conservative. A system of equations is then solved to determine the coefficients. A two-dimensional viscous Burger's equation is solved to demonstrate the conservative nature of the scheme.

Key words. conservative interpolation, overlapping grids, composite grids, finite difference methods

AMS subject classifications. 65M05, 65M50, 65N05, 65N50

1. Introduction. Standard interpolation procedures for overlapping grids do not lead to conservative methods for the solution of conservation laws. To obtain conservative schemes, it is necessary to interpolate fluxes at the overlapping boundaries instead of interpolating solution values. In one dimension we show that any consistent interpolation scheme for the fluxes is conservative. In two or three dimensions the result is not as easy. We describe a scheme to interpolate conservatively in two or three space dimensions. We implement the method in two dimensions, on a viscous Burger's equation, and show results of shocks passing through interfaces between grids. These computations demonstrate the conservative nature of the interpolation.

We are interested in numerically solving a system of conservation laws using a finite-difference or finite-volume method on overlapping grids. An overlapping grid consists of a set of logically rectangular curvilinear grids that cover a domain and overlap where they meet. Grid functions defined on the domain are matched by interpolation at the overlapping boundaries between component grids. For solving problems that are smoothly represented on the grid, standard interpolation procedures result in accurate solutions to partial differential equations (PDEs), as shown, for example, in Chesshire and Henshaw [7]. In this case it is not necessary that the discrete scheme be exactly conservative. If one is willing to have a fine enough grid to resolve sharp features such as shocks, then standard interpolation will give accurate results. For solving problems whose solution is not smoothly represented on the grid, however, it may be necessary to interpolate in a conservative way to ensure that the discrete solution converges to a physically relevant solution and to ensure that the speeds and locations of shocks are correct. For many applications it is not feasible with current computers to have enough grid points to resolve shocks. Thus we are led to develop conservative schemes. Note that, in principle, conservative interpolation is required only where shocks actually cross the boundary between grids. In many cases the problem can be avoided by aligning grids with the shocks. In general, however, it is difficult to avoid having a shock cross a grid boundary, especially in three dimensions.

An excellent discussion of conservative interpolation can be found in the paper of M. Berger [2]. She is concerned with both spatial and timewise discontinuities, the latter occurring when the timestep is different on different grids. In one space dimension Berger shows how to interpolate the fluxes in a conservative fashion given any discrete approximation to the integral. In two dimensions she shows how to interpolate conservatively for a restricted

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class of problems, when the overlapping grids are related to each other in particular ways. Although the paper suggests an algorithm for the general case, the details are somewhat lacking, and it was apparently never implemented. Also of interest is the work of M. Rai [14] who shows how to interpolate conservatively on patched grids. These grids are composite grids whose boundaries match exactly on certain curves. Conservative interpolation in this case is somewhat simpler since the boundary across which the flux must be matched is well defined and the interpolation is in one dimension lower than that of the PDE.

A conservative difference scheme exactly conserves some discrete approximation to the integral of the solution, up to contributions of fluxes through the boundaries. The conservation property must be local in the sense that the scheme should be conservative on any subdomain. For an overlapping grid the approximation to the integral is not clearly defined where the grids overlap. In general, the method of Berger's paper consists of starting with a discrete approximation to the integral and then determining the interpolation coefficients. In the approach presented here we leave both the weights in the approximation to the integral and the coefficients in the interpolation of the fluxes as free parameters. The condition that the scheme should be locally conservative imposes constraints on these parameters. In this paper we extend Berger's one-dimensional result to show that any interpolation of the fluxes is conservative. In two dimensions we describe and implement a general algorithm for conservative interpolation on overlapping grids, such as those created by the program CMPGRD [10], [5], [7].

We present numerical examples for the two-dimensional Burger's equation that show a (viscous) shock moving through an overlapping grid. The conservative interpolation weights that we generate do indeed conserve the solution to roundoff errors. In the future we will study the effectiveness of this approach to conservative interpolation when coupled with some of the state of the art methods for computing solutions to the Euler or Navier–Stokes equations. For now we remark that results from our preliminary investigations suggest that some care will be required in this regard. It is important that interpolated fluxes are independent of the local coordinate system of a component grid so that it makes sense to use the fluxes on a different component grid. Further details and numerical experiments can be found in [8].

One of our motivations for doing this work was the fact that it has been reported, by Pärt-Enander and Sjögreen [13], for example, that very slowly moving shocks can become stuck or distorted at the boundary between two overlapping grids. Whether this difficulty is solely due to the interpolation or to a coupling between the interpolation and the integration scheme is difficult to decide. Hopefully with the availability of a procedure for conservative interpolation it will be easier to isolate the source of the problem in such situations. It should be pointed out, however, that the problem only seems to occur for extremely slow moving shocks. Many researchers have successfully used nonconservative interpolation on overlapping grids to compute complex flows in two and three space dimensions, including Berger and Oliger [3], Steger et al. [6], [17], Brown [4], and Pärt-Enander and Sjögreen [12].

We have not considered the stability properties of time-stepping schemes combined with our conservative interpolation. Stability of difference schemes on overlapping grids has been considered by a number of authors including Starius [16], Reyna [15], Trefethen [19], Berger [1], and Thuné [18]. In practice we have not encountered problems due to instabilities at the overlapping boundaries.

1.1. Weak solutions and conservation. We first review some well-known concepts about conservation laws and weak solutions that will motivate our approach for developing a conservative interpolation scheme for overlapping grids. Consider the solution of the conservation equations

$$(1) \quad \mathbf{u}_t + \mathbf{f}_x + \mathbf{g}_y = 0.$$

We have in mind a nonlinear hyperbolic system such as the equations of gas dynamics. Solutions to this equation may develop discontinuities such as shocks. A *solution* with a shock is not a classical solution because it is not differentiable, but it is still physically relevant in the sense that it is the limiting behaviour as certain neglected quantities such as viscosity or heat conduction tend to zero. Thus we wish to extend our notion of solution to this more general class of solutions, called weak solutions. Typically such equations are derived from the consideration of the conservation of some quantity such as mass, momentum, or energy through a small control volume ΔV

$$(2) \quad \frac{d}{dt} \int_{\Delta V} \mathbf{u} \, d\mathbf{x} + \int_{\partial \Delta V} (\mathbf{f}, \mathbf{g}) \cdot \mathbf{n} \, ds = 0,$$

and state that the time rate of change of the integral of $\mathbf{u}$ over ΔV is equal to the integral of the normal component of the fluxes through the boundary $\partial \Delta V$. The PDE formulation follows by letting the size of this arbitrary control volume go to zero. The equivalence of the two formulations requires certain smoothness conditions on the solution. We can extend our class of solutions by looking for solutions that satisfy (2) for all possible control volumes. This formulation does not require $(\mathbf{f}, \mathbf{g})$ to be differentiable. For later reference we define two types of weak solutions.

DEFINITION 1.1. *We say that $\mathbf{u}$ is a weak solution of type I to the conservation equations (1) if it satisfies the integral form of the equations (2) for any (suitably smooth) control volume ΔV .*

The finite-volume method is the discrete version of this formulation. This method imposes a discrete form of (2) for each cell in the computational grid. There is another common way of defining a more general class of solutions.

DEFINITION 1.2. *We say that $\mathbf{u}$ is a weak solution of type II to the conservation equations (1) if*

$$(3) \quad \int_t \int_{\Omega} (\phi_t \mathbf{u} + \nabla \phi \cdot (\mathbf{f}, \mathbf{g})) \, d\mathbf{x} \, dt - \int_{\partial \Omega} \mathbf{u}(\mathbf{x}, 0) \phi(\mathbf{x}, 0) d\mathbf{x} = 0$$

for all smooth test functions $\phi(\mathbf{x}, t)$ of compact support.

This expression can be formally obtained by multiplying the conservation law by ϕ and integrating by parts. The main point is that the possibly discontinuous functions $\mathbf{u}$ and $\mathbf{f}$ are not differentiated in this formulation.

1.2. Conservative schemes on a rectangular grid. Let us now consider the numerical solution of (1) on a single rectangular grid. A discrete scheme is said to be in conservation form if it can be written as

$$(4) \quad \frac{d}{dt} \mathbf{u}_{ij} + \frac{1}{J_{ij}} (\Delta_{+i} \mathbf{F}_{ij} + \Delta_{+j} \mathbf{G}_{ij}) = 0,$$

where

$$\begin{aligned} J_{ij} &= \text{approximate area of cell } ij, \\ \mathbf{F}_{ij} &= \text{approximation to } \mathbf{f} \text{ on left face of cell } ij, \\ \mathbf{G}_{ij} &= \text{approximation to } \mathbf{g} \text{ on bottom face of cell } ij, \end{aligned}$$

and the difference operators are defined by

$$\begin{aligned} \Delta_{+i} \mathbf{F}_{ij} &= \mathbf{F}_{i+1,j} - \mathbf{F}_{ij}, & \Delta_{+j} \mathbf{G}_{ij} &= \mathbf{G}_{ij+1} - \mathbf{G}_{ij}, \\ \Delta_{-i} \mathbf{F}_{ij} &= \mathbf{F}_{ij} - \mathbf{F}_{i-1,j}, & \Delta_{-j} \mathbf{G}_{ij} &= \mathbf{G}_{ij} - \mathbf{G}_{ij-1}. \end{aligned}$$

In particular the finite-volume method of discretization generates a scheme in conservation form. If we consider any subdomain $B \subset \Omega$, then a scheme in conservation form conserves an approximation to the integral of $\mathbf{u}$ over B

$$S_B(t) = \sum_{ij \in B} \mathbf{u}_{ij} J_{ij}$$

except for the fluxes coming through ∂B , the boundary of B . To see this, simply take the time derivative of S_B and substitute (4)

$$\begin{aligned} \frac{dS_B}{dt} &= \sum_{ij \in B} \frac{d\mathbf{u}_{ij}}{dt} J_{ij} = - \sum_{ij \in B} \{ \Delta_{+i} \mathbf{F}_{ij} + \Delta_{+j} \mathbf{G}_{ij} \} \\ &= \sum_{ij \in \partial B} \pm \mathbf{F}_{ij} \pm \mathbf{G}_{ij} \\ &= \text{fluxes crossing the boundary.} \end{aligned}$$

All of the fluxes in the interior of B cancel, leaving only boundary terms. In the discrete case we see that the equation in conservation form satisfies the discrete analogue of the weak solution of type I (Def. 1.1).

Suppose that we have a discrete approximation to the integral and a discrete scheme in conservation form. Multiplication of the discrete scheme by $\phi_{ij} J_{ij}$ and integrating (summing) gives

$$(5) \quad \int_t \sum_{ij} \left\{ \phi_{ij} \frac{d}{dt} \mathbf{u}_{ij} J_{ij} + \phi_{ij} (\Delta_{+i} \mathbf{F}_{ij} + \Delta_{+j} \mathbf{G}_{ij}) \right\} dt = 0.$$

Summation by parts gives

$$(6) \quad \begin{aligned} \int_t \sum_{ij} \left\{ \frac{d}{dt} \phi(t)_{ij} \mathbf{u}_{ij} + \frac{1}{J_{ij}} \{ (\Delta_{-i} \phi_{ij}) \mathbf{F}_{ij} + (\Delta_{-j} \phi_{ij}) \mathbf{G}_{ij} \} \right\} J_{ij} dt \\ - \sum_{ij} \phi(0)_{ij} \mathbf{u}(0)_{ij} J_{ij} = 0. \end{aligned}$$

Since ϕ is smooth, if the solution converges then the above sums will converge to the integrals appearing in the definition of a weak solution of type II. In this case the solution will be a weak solution of type II. Thus we see that a scheme in conservation form, a discrete version of the integrated form of the conservation laws, (2), is compatible with the definition of weak solution type II (Def. 1.2). This idea that a solution to the equations in conservation form will converge to a weak solution of type II, provided the solution converges at all, is due to Lax and Wendroff [11].

It is important to understand that a conservative scheme must have the local conservation property. The integral (or approximation to the integral) over any subdomain must be conserved up to the fluxes crossing the boundary of the subdomain. It is not enough that the integral over the entire domain be conserved since this alone does not ensure that the solution will be a weak solution. In particular, shocks may travel with the wrong speed. Thus, for example, the naive approach of adjusting all values by a fixed amount at each timestep to be globally conservative is not locally conservative. This naive approach is not wrong in itself, but rather it is a mistake to believe that it will generate a weak solution or that shocks must necessarily move with the correct speed.

2. Conservation on one-dimensional overlapping grids. We consider the solution of the conservation equation

$$\mathbf{u}_t + \mathbf{f}_x = 0$$

on a one-dimensional overlapping grid on the interval $[a, b]$; see Fig. 1.

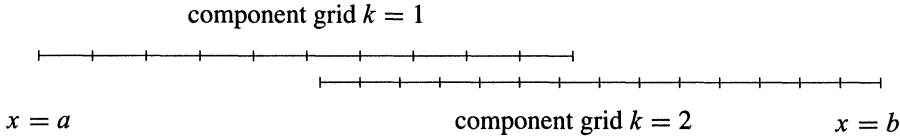


FIG. 1. One-dimensional overlapping grid for the interval $[a, b]$.

Solutions to this equation conserve the integral

$$I(t) = \int_a^b \mathbf{u}(x, t) dx.$$

That is,

$$\frac{dI}{dt} = \int_a^b \frac{\partial \mathbf{u}}{\partial t} dx = - \int_a^b \mathbf{f}_x dx = -[\mathbf{f}]_a^b.$$

The overlapping grid has two component grids labelled by $k = 1, 2$. The grid points on grid k are numbered $i = 0, \dots, N_k + 1$; see Fig. 2.

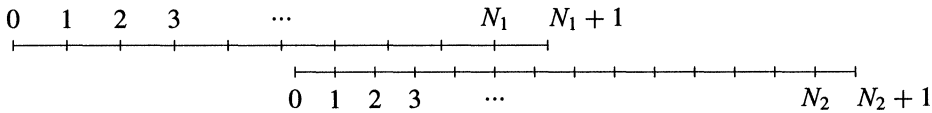


FIG. 2. Numbering of grid points.

Let $\mathbf{u}_{k,i}$ denote the discrete approximation to $\mathbf{u}$ on component grid $k = 1, 2$ for $i = 0, \dots, N_k + 1$, defined at the midpoints of the cells and let $\mathbf{f}_{k,i}$ denote the discrete approximation to $\mathbf{f}$ on the cell edges; see Fig. 3.

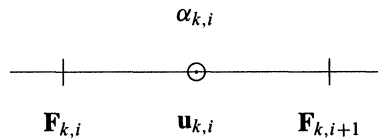


FIG. 3. Location of discrete variables.

The grid points will be denoted by $x_{k,i}$ with grid spacings $h_{k,i} = x_{k,i+1} - x_{k,i}$. Define the undivided difference operators

$$\Delta_+ \mathbf{F}_{k,i} = \mathbf{F}_{k,i+1} - \mathbf{F}_{k,i}, \quad \Delta_- \mathbf{F}_{k,i} = \mathbf{F}_{k,i} - \mathbf{F}_{k,i-1},$$

and introduce the discrete scheme

$$(7) \quad \frac{d\mathbf{u}_{k,i}}{dt} + \frac{1}{h_{k,i}} \Delta_+ \mathbf{F}_{k,i} = 0$$

on the two component grids $k = 1, 2$. Here the numerical flux $\mathbf{F}_{k,i}$ is some function of $\mathbf{u}_{k,i}$

$$\mathbf{F}_{k,i} = \mathbf{F}_N(\dots, \mathbf{u}_{k,i-1}, \mathbf{u}_{k,i}, \mathbf{u}_{k,i+1}, \dots)$$

such that $\mathbf{F}_N(\dots, \mathbf{u}, \mathbf{u}, \mathbf{u}, \dots) = \mathbf{f}(\mathbf{u})$. On an overlapping grid, extra information is needed on the boundary of a component grid that overlaps another component grid. The standard approach is to interpolate the solution value at such an *interpolation point* in terms of the solution values on the grid that is being overlapped. However, to achieve a conservative method, the flux is interpolated. (This comes from the fact that the flux function appears in a linear fashion in the conservation law while $\mathbf{u}$ has, in general, a nonlinear dependence.) The interpolation conditions are of the form (see Fig. 4)

$$(8) \quad \mathbf{F}_{1,N_1+1} = \sum_{j=M_1}^{M_2} \gamma_j \mathbf{F}_{2,j} \quad 0 \leq M_1 \leq M_2 \leq N_2,$$

$$(9) \quad \mathbf{F}_{2,0} = \sum_{j=L_1}^{L_2} \beta_j \mathbf{F}_{1,j} \quad 0 \leq L_1 \leq L_2 \leq N_1.$$

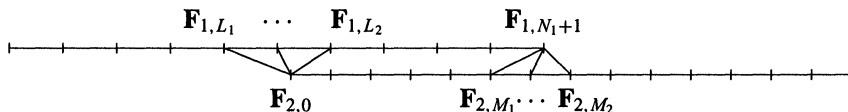


FIG. 4. Interpolation of endpoints.

Introduce the discrete approximation to the integral $I(t)$,

$$S(t) = \sum_{k=1}^2 \sum_{i=0}^{N_k} \alpha_{k,i} \mathbf{u}_{k,i} h_{k,i}.$$

We will show that in one dimension, for any consistent interpolation of the fluxes (8), (9), the scheme is conservative. We first show that the scheme satisfies the discrete analogue of the local conservation property (Def. 1.1) since the algebra is somewhat simpler. We then show that the scheme is also consistent with the definition of a weak solution of type II.

THEOREM 2.1. *If the interpolation coefficients are first-order accurate,*

$$\sum_i \beta_i = 1 \quad \text{and} \quad \sum_i \gamma_i = 1,$$

and are bounded,

$$\sum_i |\beta_i| = O(1) \quad \text{and} \quad \sum_i |\gamma_i| = O(1),$$

then there exist $\alpha_{k,i}$ so that $dS/dt = -[\mathbf{f}]_a^b$ for any interval $[a, b]$ and $S(t) = I(t) + O(h)$, where $h = \max_{k,i} h_{k,i}$. If the interpolation is second-order accurate, then the integral is approximated to second order: $S(t) = I(t) + O(h^2)$. Note that the $\alpha_{k,i}$ must be chosen independent of $\mathbf{u}_k$ and $\mathbf{F}_k$.

Proof. It holds that

$$\frac{dS}{dt} = \sum_{i=0}^{N_1} \alpha_{1,i} \frac{d\mathbf{u}_{1,i}}{dt} h_{1,i} + \sum_{i=0}^{N_2} \alpha_{2,i} \frac{d\mathbf{u}_{2,i}}{dt} h_{2,i}$$

$$\begin{aligned}
 &= -\sum_{i=0}^{N_1} \alpha_{1,i} \Delta_+ \mathbf{F}_{1,i} - \sum_{i=0}^{N_2} \alpha_{2,i} \Delta_+ \mathbf{F}_{2,i} \\
 &= \alpha_{1,0} \mathbf{F}_{1,0} + \sum_1^{N_1} \Delta_-(\alpha_{1,i}) \mathbf{F}_{1,i} - \alpha_{1,N_1} \mathbf{F}_{1,N_1+1} \\
 &\quad + \alpha_{2,0} \mathbf{F}_{2,0} + \sum_1^{N_2} \Delta_-(\alpha_{2,i}) \mathbf{F}_{2,i} - \alpha_{2,N_2} \mathbf{F}_{2,N_2+1}.
 \end{aligned}$$

Using (8) and (9) for $\mathbf{F}_{1,N_1+1}$ and $\mathbf{F}_{2,0}$ gives

$$\begin{aligned}
 \frac{dS}{dt} &= \alpha_{1,0} \mathbf{F}_{1,0} + \sum_1^{L_1-1} \Delta_-(\alpha_{1,i}) \mathbf{F}_{1,i} + \sum_{L_1}^{L_2} \{ \Delta_-(\alpha_{1,i}) + \alpha_{2,0} \beta_i \} \mathbf{F}_{1,i} \\
 &\quad + \sum_{L_2+1}^{N_2} \Delta_-(\alpha_{1,i}) \mathbf{F}_{1,i} + \sum_1^{M_1-1} \Delta_-(\alpha_{2,i}) \mathbf{F}_{2,i} \\
 &\quad + \sum_{M_1}^{M_2} \{ \Delta_-(\alpha_{2,i}) - \alpha_{1,N_1} \gamma_i \} \mathbf{F}_{2,i} \\
 &\quad + \sum_{M_2+1}^{N_2} \Delta_-(\alpha_{2,i}) \mathbf{F}_{2,i} - \alpha_{2,N_2} \mathbf{F}_{2,N_2+1}.
 \end{aligned}$$

We now impose the conservation condition, $dS/dt = \mathbf{f}(a) - \mathbf{f}(b)$ for all $\mathbf{F}_{k,i}$. For simplicity, we assume that the boundary conditions are $\mathbf{F}_{1,0} = \mathbf{f}(a)$ and $\mathbf{F}_{2,N_2+1} = \mathbf{f}(b)$, so

$$\begin{aligned}
 \alpha_{1,0} &= 1, \\
 \alpha_{2,N_2} &= 1, \\
 \Delta_-(\alpha_{1,i}) &= 0 \quad i = 1, \dots, L_1 - 1, L_2 + 1, \dots, N_1, \\
 \Delta_-(\alpha_{1,i}) + \alpha_{2,0} \beta_i &= 0 \quad i = L_1, \dots, L_2, \\
 \Delta_-(\alpha_{2,i}) &= 0 \quad i = 1, \dots, M_1 - 1, M_2 + 1, \dots, N_2, \\
 \Delta_-(\alpha_{2,i}) - \alpha_{2,0} \gamma_i &= 0 \quad i = M_1, \dots, M_2.
 \end{aligned}$$

These equations are equivalent to the following set of equations, where we have defined $\mu = \alpha_{2,0}$ and have used the fact that $\sum_j \beta_j = 1$ and $\sum_j \gamma_j = 1$:

$$\begin{aligned}
 \alpha_{1,i} &= 1 & i = 0, \dots, L_1 - 1, \\
 \alpha_{1,i} &= \alpha_{1,i-1} - \mu \beta_i & i = L_1, \dots, L_2 - 1, \\
 \alpha_{1,i} &= 1 - \mu & i = L_2, \dots, N_1, \\
 \alpha_{2,i} &= \mu & i = 0, \dots, M_1 - 1, \\
 \alpha_{2,i} &= \alpha_{2,i-1} + (1 - \mu) \gamma_i & i = M_1, \dots, M_2 - 1, \\
 \alpha_{2,i} &= 1 & i = M_2, \dots, N_2.
 \end{aligned} \tag{10}$$

These last equations define the general solution to the problem in terms of the one free parameter μ . Thus we see that away from the overlap region and interpolation regions the integration weights $\alpha_{k,i}$ are equal to 1. In the region of overlap but away from the interpolation regions, the weights are also constant, with $\alpha_{1,i} = 1 - \mu$ and $\alpha_{2,i} = \mu$ (see Fig. 5).

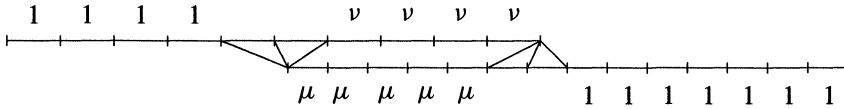


FIG. 5. Solution for the integration weights $\alpha_{k,i}$. The weights are constant outside the interpolation regions. In the overlap region one grid has $\alpha_{1,i} = \nu$ and the other has $\alpha_{2,i} = \mu$ with $\mu + \nu = 1$.

The parameter μ defines how to weight the two component grids in the sum $S(t)$ in the region where they overlap. Component grid $k = 1$ is weighted by $1 - \mu$ and component grid $k = 2$ is weighted by μ . Taking $\mu = \frac{1}{2}$ would give equal weight to each component grid in its contribution to the integral, although note that the numerical scheme (7), (8), (9) is independent of μ .

The equations (10) define a set of constraints on the interpolation coefficients γ_i , β_i and the weights α_i in the discrete approximation to the integral that must be satisfied for the scheme to be conservative. Provided we can choose coefficients that are consistent with these constraints and that lead to accurate interpolation and integration formulae then we will have a conservative scheme. In higher dimensions a similar result holds, although the constraints become much more complicated.

To complete the proof of the theorem it remains to show that $S(t)$ approximates the integral of $\mathbf{u}$, for smooth functions $\mathbf{u}$.

$$\begin{aligned} S(t) &= \sum_{k=1}^2 \sum_{i=0}^{N_k} \alpha_{k,i} \mathbf{u}_{k,i} h_{k,i} \\ &= \sum_0^{L_1-1} \mathbf{u}_{1,i} h_{1,i} + \sum_{L_1}^{L_2-1} \alpha_{1,i} \mathbf{u}_{1,i} h_{1,i} + \sum_{L_2}^{N_1} (1 - \mu) \mathbf{u}_{1,i} h_{1,i} \\ &\quad + \sum_0^{M_1-1} \mu \mathbf{u}_{2,i} h_{2,i} + \sum_{M_1}^{M_2-1} \alpha_{2,i} \mathbf{u}_{2,i} h_{2,i} + \sum_{M_2}^{N_2} \mathbf{u}_{2,i} h_{2,i}. \end{aligned}$$

The sums that apply to points outside the interpolation regions approximate integrals over the appropriate intervals:

$$\begin{aligned} S(t) &= \int_a^{x_{1,L_1}} \mathbf{u} \, dx + \sum_{L_1}^{L_2-1} \alpha_{1,i} \mathbf{u}_{1,i} h_{1,i} + (1 - \mu) \int_{x_{1,L_2}}^{x_{1,N_1+1}} \mathbf{u} \, dx \\ &\quad + \mu \int_{x_{2,0}}^{x_{2,M_1}} \mathbf{u} \, dx + \sum_{M_1}^{M_2-1} \alpha_{2,i} \mathbf{u}_{2,i} h_{2,i} + \int_{x_{2,M_2}}^b \mathbf{u} \, dx + O(h^2), \end{aligned}$$

where $x_{k,i}$ is the position of the i th grid point on grid k . Consider the sum over the interpolation interval $[L_1, L_2 - 1]$ on component grid 1

$$S_1(L_1, L_2 - 1) = \sum_{i=L_1}^{L_2-1} \alpha_{1,i} \mathbf{u}_{1,i} h_{1,i}.$$

The variables involved in the interpolation are shown in Fig. 6.

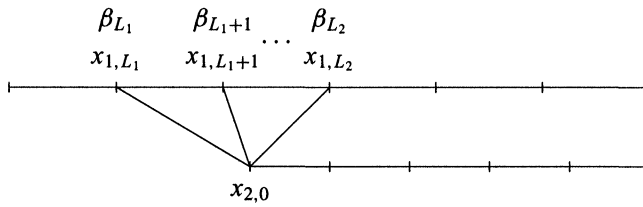


FIG. 6. Grid variables in the interpolation region.

Since

$$\alpha_{1,i} = 1 - \mu(\beta_{L_1} + \beta_{L_1+1} + \dots + \beta_i) \quad \text{for } i = L_1, \dots, L_2 - 1;$$

therefore

$$S_1(L_1, L_2 - 1) = \sum_{i=L_1}^{L_2-1} \mathbf{u}_{1,i} h_{1,i} - \mu \sum_{i=L_1}^{L_2-1} \left\{ \sum_{j=L_1}^i \beta_j \right\} \mathbf{u}_{1,i} h_{1,i}.$$

The first sum in this last expression approximates the integral of $\mathbf{u}$ from x_{1,L_1} to x_{1,L_2} . Changing the order of summation in the second sum gives

$$\sum_{i=L_1}^{L_2-1} \left\{ \sum_{j=L_1}^i \beta_j \right\} \mathbf{u}_{1,i} h_{1,i} = \sum_{i=L_1}^{L_2-1} \beta_i \left\{ \sum_{j=i}^{L_2-1} \mathbf{u}_{1,j} h_{1,j} \right\} = \sum_{i=L_1}^{L_2} \beta_i \int_{x_{1,i}}^{x_{1,L_2}} \mathbf{u} \, dx + O(h^2).$$

The β_i were chosen to interpolate discrete functions on the grid points $x_{L_1}, \dots, x_{L_2}$ to obtain a value at $x_{2,0}$. In this last expression we recognize that we are interpolating the function that is equal to the integral of $\mathbf{u}$ from x to x_{L_2} and thus

$$\sum_{i=L_1}^{L_2-1} \left\{ \sum_{j=L_1}^i \beta_j \right\} \mathbf{u}_{1,i} h_{1,i} = \int_{x_{2,0}}^{x_{1,L_2}} \mathbf{u} \, dx + O(h^p),$$

where p is 1 or 2 depending on the order of accuracy of the interpolation and whence

$$S_1(L_1, L_2 - 1) = \int_{x_{1,L_1}}^{x_{1,L_2}} \mathbf{u} \, dx - \mu \int_{x_{2,0}}^{x_{1,L_2}} \mathbf{u} \, dx + O(h^p).$$

Similarly we estimate the sum $S_2(M_1, M_2 - 1)$

$$S_2(M_1, M_2 - 1) = \int_{x_{2,M_1}}^{x_{2,M_2}} \mathbf{u} \, dx - (1 - \mu) \int_{x_{2,M_1}}^{x_{1,N_1+1}} \mathbf{u} \, dx + O(h^p).$$

It follows that

$$S(t) = \int_a^b \mathbf{u} \, dx + O(h^p). \quad \square$$

The scheme that we have constructed for one-dimensional overlapping grids satisfies the discrete local conservation property consistent with the definition of a weak solution of type I. We now show that solutions to this scheme will also converge to weak solutions of type II.

THEOREM 2.2. *If the discrete scheme for one-dimensional overlapping grids (7,8,9) satisfies the assumptions of Theorem 2.1 and the discrete solution converges as the mesh is refined, then the discrete solution converges to a weak solution of type II (Def. 1.2).*

Proof. Multiplying the discrete scheme, (7), by $\phi_{k,i} h_{k,j}$ and integrating (summing) gives

$$\int_t \sum_{k,i} \alpha_{k,i} \left\{ \phi_{k,i} \frac{d}{dt} \mathbf{u}_{k,i} h_{k,i} + \phi_{k,i} \Delta_{+i} \mathbf{F}_{k,i} \right\} dt = 0.$$

Integration by parts in time and summation by parts in space (proceeding in the same fashion as in the proof of Theorem 2.1) gives

$$\begin{aligned} & \sum_{k,i} \phi_{k,i}(t_0) \mathbf{u}_{k,i}(t_0) h_{k,i} - \int_t \sum_{k,i} \frac{d\phi_{k,i}}{dt} \mathbf{u}_{k,i} h_{k,i} dt \\ & - \int_t \left\{ \alpha_{1,0} \phi_{1,0} \mathbf{F}_{1,0} + \sum_1^{L_1-1} \Delta_{-}(\alpha_{1,i} \phi_{1,i}) \mathbf{F}_{1,i} + \sum_{L_1}^{L_2} \left\{ \Delta_{-}(\alpha_{1,i} \phi_{1,i}) + \alpha_{2,0} \phi_{2,0} \beta_i \right\} \mathbf{F}_{1,i} \right. \\ & \quad + \sum_{L_2+1}^{N_2} \Delta_{-}(\alpha_{1,i} \phi_{1,i}) \mathbf{F}_{1,i} + \sum_1^{M_1-1} \Delta_{-}(\alpha_{2,i}) \phi_{2,i} \mathbf{F}_{2,i} \\ & \quad + \sum_{M_1}^{M_2} \left\{ \Delta_{-}(\alpha_{2,i} \phi_{2,i}) - \alpha_{1,N_1} \phi_{1,N_1} \gamma_i \right\} \mathbf{F}_{2,i} + \sum_{M_2+1}^{N_2} \Delta_{-}(\alpha_{2,i} \phi_{2,i}) \mathbf{F}_{2,i} \\ & \quad \left. - \alpha_{2,N_2} \phi_{2,N_2} \mathbf{F}_{2,N_2+1} \right\} dt = 0. \end{aligned}$$

Using $\Delta_{-}(f_i g_i) = \Delta_{-}(f_i) g_i + f_{i-1} \Delta_{-}(g_i)$ gives

$$\begin{aligned} \Delta_{-}(\alpha_{1,i} \phi_{1,i}) + \alpha_{2,0} \phi_{2,0} \beta_i &= \left\{ \Delta_{-}(\alpha_{1,i}) + \alpha_{2,0} \beta_i \right\} \phi_{1,i} \\ &\quad + \alpha_{1,i-1} \Delta_{-}(\phi_{1,i}) + \alpha_{2,0} \beta_i (\phi_{2,0} - \phi_{1,i}) \\ &= \alpha_{1,i-1} \Delta_{-}(\phi_{1,i}) + \alpha_{2,0} \beta_i (\phi_{2,0} - \phi_{1,i}). \end{aligned}$$

Define

$$I(L_1, L_2) = \sum_{L_1}^{L_2} \alpha_{2,0} \beta_i (\phi_{2,0} - \phi_{1,i}) \mathbf{F}_{1,i}$$

and note that for smooth functions ϕ of compact support, $\phi_{2,0} - \phi_{1,i} = O(h)$ and thus

$$\int_t I(L_1, L_2) dt = O(h)$$

provided $|L_2 - L_1| = O(1)$, $\alpha_{2,0} = O(1)$, and $\beta_i = O(1)$. If $\mathbf{F}_{1,i}$ converges to a smooth function then $I(L_1, L_2) = O(h^p)$, but generally we can only assume that $\mathbf{F}_{1,i}$ is bounded. Collecting terms, we get

$$\begin{aligned} & \sum_{k,i} \phi_{k,i}(t_0) \mathbf{u}_{k,i}(t_0) h_{k,i} - \int_t \sum_{k,i} \frac{d\phi_{k,i}}{dt} \mathbf{u}_{k,i} h_{k,i} dt \\ & - \int_t \left\{ \alpha_{1,0} \phi_{1,0} \mathbf{F}_{1,0} + \sum_1^{N_1} \alpha_{1,i} \Delta_{-}(\phi_{1,i}) \mathbf{F}_{1,i} + \sum_1^{N_2} \alpha_{2,i} \Delta_{-}(\phi_{2,i}) \mathbf{F}_{2,i} \right. \\ & \quad \left. - \alpha_{2,N_2} \phi_{2,N_2} \mathbf{F}_{2,N_2+1} \right\} dt + O(h) = 0. \end{aligned}$$

Since the test function ϕ is smooth (and of compact support so that the boundary terms disappear) and since the α_i were chosen so the sum converges to the integral, this final expression converges to

$$\int_t \int_x (\phi_t \mathbf{u} + \phi_x \mathbf{F}) dx dt + \int_x \phi(x, t_0) \mathbf{u}(x, t_0) dx = 0$$

and thus the discrete solution converges to a weak solution of type II. $\square$

3. Conservation in two dimensions. We now describe how to interpolate in a conservative way in two space dimensions. The algorithm we describe generalizes easily to three space dimensions. In two dimensions we consider the solution of the conservation equation

$$(11) \quad \mathbf{u}_t + \mathbf{f}_x + \mathbf{g}_y = 0$$

on a domain Ω . The equation is conservative in the sense that the integral

$$I(t) = \int_{\Omega} \mathbf{u}(x, y, t) dx dy$$

has the property that

$$\frac{dI(t)}{dt} = \int_{\Omega} \frac{\partial \mathbf{u}}{\partial t} dx dy = - \int_{\Omega} (\mathbf{f}_x + \mathbf{g}_y) dx dy = - \int_{\partial \Omega} (\mathbf{f}, \mathbf{g}) \cdot \mathbf{n} ds.$$

The domain Ω is discretized using overlapping grids as described in Chesshire and Henshaw [7]. The domain is completely covered by a set of curvilinear grids. For each component grid $k = 1, 2, \dots, n_g$ there is a mapping

$$\mathbf{d}_k : [0, 1]^2 \rightarrow \Omega_k$$

from the unit square to domain Ω_k , $\mathbf{x} = \mathbf{d}_k(\mathbf{r}) = (x_k(\mathbf{r}), y_k(\mathbf{r}))$. It will be convenient to introduce a related mapping for which the grid spacing in each direction will be one. Suppose component grid k is discretized with N_r and N_s grid lines in the r and s directions. Then define

$$\mathbf{D}_k(\mathbf{R}) = \mathbf{d}_k(R/(N_r - 1), S/(N_s - 1))$$

where

$$\mathbf{R} = (R, S), \quad R \in [0, N_r - 1], \quad S \in [0, N_s - 1].$$

The Jacobian of this mapping is

$$J_k = \left| \frac{\partial \mathbf{D}_k}{\partial \mathbf{R}} \right| = \frac{\partial x_k}{\partial R} \frac{\partial y_k}{\partial S} - \frac{\partial x_k}{\partial S} \frac{\partial y_k}{\partial R}.$$

We transform the conservation equation (11) on each component grid to

$$(12) \quad \mathbf{u}_t + \frac{1}{J_k} \left(\frac{\partial \mathbf{F}_k}{\partial R} + \frac{\partial \mathbf{G}_k}{\partial S} \right) = 0,$$

where

$$(13) \quad \begin{aligned} (\mathbf{F}_k, \mathbf{G}_k) &= \left(\frac{\partial y_k}{\partial S} \mathbf{f} - \frac{\partial x_k}{\partial S} \mathbf{g}, -\frac{\partial y_k}{\partial R} \mathbf{f} + \frac{\partial x_k}{\partial R} \mathbf{g} \right) \\ (\mathbf{f}, \mathbf{g}) &= \frac{1}{J_k} \left(\frac{\partial x_k}{\partial R} \mathbf{F} + \frac{\partial x_k}{\partial S} \mathbf{G}, \frac{\partial y_k}{\partial R} \mathbf{F} + \frac{\partial y_k}{\partial S} \mathbf{G} \right). \end{aligned}$$

Let $\mathbf{u}_{k,ij}$ denote the grid function values on grid k . The $\mathbf{u}_{k,ij}$ is cell-centred, while $\mathbf{F}_{k,ij}$ and $\mathbf{G}_{k,ij}$ is face-centred (Fig. 7).

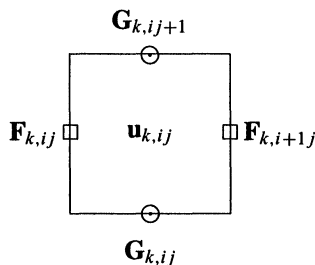


FIG. 7. Cell variables in two dimensions.

The discrete approximation is given by

$$\begin{aligned} \frac{d\mathbf{u}_{k,ij}}{dt} + \frac{1}{J_{k,ij}} (\Delta_{+i}\mathbf{F}_{k,ij} + \Delta_{+j}\mathbf{G}_{k,ij}) &= 0, \\ \mathbf{F}_{k,ij} &= \mathbf{F}_N(\dots, \mathbf{u}_{k,i-1j}, \mathbf{u}_{k,ij}, \dots), \\ \mathbf{G}_{k,ij} &= \mathbf{G}_N(\dots, \mathbf{u}_{k,ij-1}, \mathbf{u}_{k,ij}, \dots). \end{aligned}$$

The numerical fluxes are approximated by some functions $\mathbf{F}_N$ and $\mathbf{G}_N$ (N stands for numerical approximation) which for consistency should satisfy

$$\mathbf{F}_N(\dots, \mathbf{u}, \mathbf{u}, \dots) = \mathbf{F}_k(\mathbf{u}) \quad \text{and} \quad \mathbf{G}_N(\dots, \mathbf{u}, \mathbf{u}, \dots) = \mathbf{G}_k(\mathbf{u}).$$

To completely specify the discrete scheme, we need some appropriate boundary conditions at true boundaries and some interpolation conditions at the interpolation boundaries. The latter will be specified shortly.

Introduce the discrete approximation to the integral

$$S(t) = \sum_{k,ij} \alpha_{k,ij} \mathbf{u}_{k,ij} J_{k,ij}.$$

Note that by the definition of R and S , the Jacobian $J_{k,ij}$ will be approximately the area of the cell. Taking the time derivative of $S(t)$ and substituting the conservation equations gives

$$\frac{dS}{dt} = \sum_{k,ij} \alpha_{k,ij} \frac{d\mathbf{u}_{k,ij}}{dt} J_{k,ij} = - \sum_{k,ij} \alpha_{k,ij} (\Delta_{+i}\mathbf{F}_{k,ij} + \Delta_{+j}\mathbf{G}_{k,ij}).$$

Discrete integration by parts gives

$$\begin{aligned} (14) \quad \frac{dS}{dt} &= \sum_{k,ij} (\Delta_{-i}(\alpha_{k,ij})\mathbf{F}_{k,ij} + \Delta_{-j}(\alpha_{k,ij})\mathbf{G}_{k,ij}) \\ &\quad + \sum_{\partial\Omega_L} \alpha_{k,ij}\mathbf{F}_{k,ij} - \sum_{\partial\Omega_R} \alpha_{k,ij}\mathbf{F}_{k,i+1j} + \sum_{\partial\Omega_B} \alpha_{k,ij}\mathbf{G}_{k,ij} - \sum_{\partial\Omega_T} \alpha_{k,ij}\mathbf{G}_{k,ij+1}, \end{aligned}$$

where the boundary terms involve points on the Left ($\partial\Omega_L$), Right ($\partial\Omega_R$), Bottom ($\partial\Omega_B$), and Top ($\partial\Omega_T$) of the boundary cells. Some of the boundary terms will be obtained from boundary conditions on the PDE, while others will be obtained from interpolation.

We take the following approach to determine a conservative scheme.

1. All boundary terms appearing in (14) are defined using boundary conditions on the PDE or by interpolation equations. In the case of interpolation we define the flux on the interpolation boundary in terms of interior flux values on other component grids.

2. We substitute the boundary conditions and interpolation equations and collect like terms in the fluxes $(\mathbf{F}_{k,ij}, \mathbf{G}_{k,ij})$. We then set to zero the coefficients of all interior fluxes since we require that dS/dt converge to the integral of the fluxes through the boundary.

3.1. Interpolation of fluxes on overlapping grids. Certain details of the approach we take are dependent on the way in which overlapping grids are generated by the grid generation program, CMPGRD [7]. CMPGRD can generate either cell-centred or vertex-overlapping grids. For a cell-centred grid, values of $\mathbf{u}_{k,ij}$ are assumed to lie at cell-centres and interpolation is performed at cell-centres. In a vertex grid, function values lie at vertices and interpolation is done at vertices. Neither of these choices is optimal for implementing conservative interpolation since we would prefer to interpolate fluxes on cell faces. As a compromise we use a vertex grid but think of $\mathbf{u}_{k,ij}$ as being located at cell-centres. The fluxes are interpolated from the faces of one grid to the vertices of the second grid and these vertex values are then averaged to obtain face-centred values (see Fig. 8).

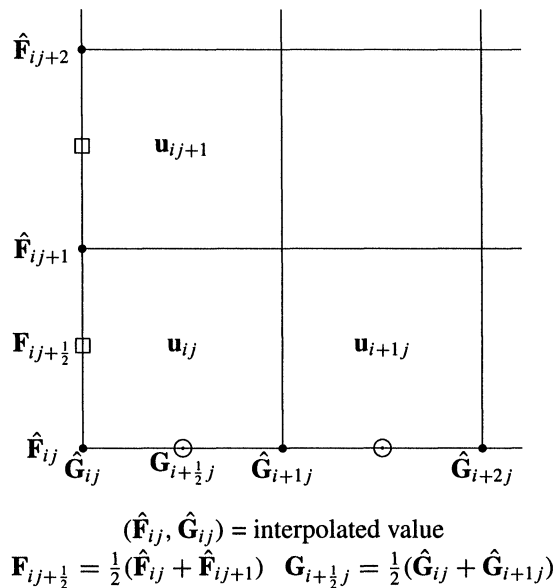


FIG. 8. Fluxes are interpolated to vertices and then averaged to obtain values on faces.

Suppose we wish to interpolate a point on component grid k from points on component grid k' . CMPGRD supplies the information needed to perform this interpolation, such as the points to be interpolated from and the (r, s) coordinates on grid k' of the point to be interpolated. The interpolation equations are of the form

$$(15) \quad \hat{\mathbf{F}}_I = \sum_{mn} \gamma_{k,ij,k',mn}^F \mathbf{F}_{k',mn}, \quad \hat{\mathbf{G}}_I = \sum_{mn} \gamma_{k,ij,k',mn}^G \mathbf{G}_{k',mn},$$

where the interpolation weights γ^F and γ^G are left unspecified for the time being (see Fig. 9).

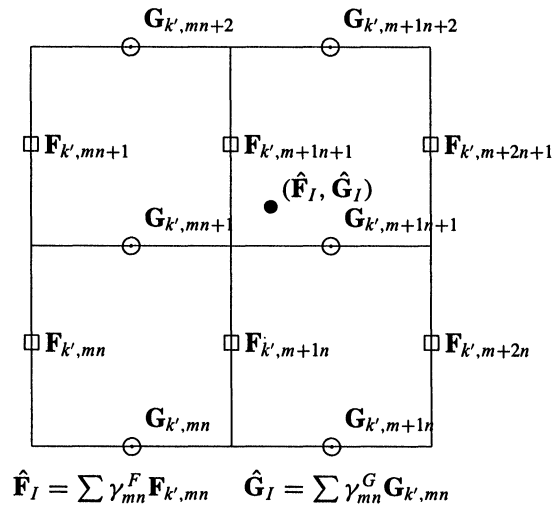


FIG. 9. The interpolated fluxes $(\hat{\mathbf{F}}_I, \hat{\mathbf{G}}_I)$ for grid k are interpolated from face-centred fluxes of grid k' .

The fluxes $(\hat{\mathbf{F}}_I, \hat{\mathbf{G}}_I)$ now have to be transformed from the coordinate system of grid k' to the coordinate system of grid k . This transformation is defined by (13). Introduce the matrix transformations

$$A_{k'}^{-1} = \frac{1}{J_{k'}} \begin{bmatrix} \partial x_{k'}/\partial R & \partial x_{k'}/\partial S \\ \partial y_{k'}/\partial R & \partial y_{k'}/\partial S \end{bmatrix}, \quad A_k = \begin{bmatrix} \partial y_k/\partial S & -\partial x_k/\partial S \\ -\partial y_k/\partial R & \partial x_k/\partial R \end{bmatrix}.$$

The interpolated vertex flux is given by

$$(16) \quad \begin{bmatrix} \hat{\mathbf{F}}_{k,ij} \\ \hat{\mathbf{G}}_{k,ij} \end{bmatrix} = A_k A_{k'}^{-1} \begin{bmatrix} \hat{\mathbf{F}}_I \\ \hat{\mathbf{G}}_I \end{bmatrix} = A_k A_{k'}^{-1} \begin{bmatrix} \sum \gamma^F \mathbf{F}_{k'} \\ \sum \gamma^G \mathbf{G}_{k'} \end{bmatrix}.$$

The matrix $A_{k'}^{-1}$ is the transformation from the fluxes in grid k' coordinates to the fluxes $(\mathbf{f}, \mathbf{g})$ in physical coordinates. The matrix A_k is the transformation matrix from the physical fluxes $(\mathbf{f}, \mathbf{g})$ to the fluxes in grid k coordinates. The matrix $A_{k'}^{-1}$ should be evaluated at the same position as $(\hat{\mathbf{F}}_I, \hat{\mathbf{G}}_I)$; thus the matrix coefficients should be interpolated from the vertices.

To obtain a flux on the face, we average the values from the two adjacent vertices obtaining either

$$(17) \quad \mathbf{F}_{k,ij+1/2} = \frac{1}{2} (\hat{\mathbf{F}}_{k,ij} + \hat{\mathbf{F}}_{k,ij+1})$$

for a vertical face or

$$(18) \quad \mathbf{G}_{k,i+1/2j} = \frac{1}{2} (\hat{\mathbf{G}}_{k,ij} + \hat{\mathbf{G}}_{k,i+1j})$$

for a horizontal face.

Let us define some sets of indices:

1. Ω_I : Set of interior cells.
2. Ω_F : set of all faces which are used to interpolate $\hat{\mathbf{F}}_I$.
3. Ω_G : set of all faces which are used to interpolate $\hat{\mathbf{G}}_I$.
4. Ω_{BF} : Points on true boundaries that require a value for $\mathbf{F}$.
5. Ω_{BG} : Points on true boundaries that require a value for $\mathbf{G}$.

Substituting the interpolation equations defined by (15)–(18) into (14) and collecting like terms in $\mathbf{F}_{k,ij}$ and $\mathbf{G}_{k,ij}$ gives

$$\begin{aligned} \frac{dS}{dt} = & \sum_{\Omega_I} (\Delta_{-i}(\alpha_{k,ij}) \mathbf{F}_{k,ij} + \Delta_{-j}(\alpha_{k,ij}) \mathbf{G}_{k,ij}) \\ & + \sum_{\Omega_F} (\Delta_{-i}(\alpha_{k,ij}) - \beta_{k,ij}^F) \mathbf{F}_{k,ij} + \sum_{\Omega_G} (\Delta_{-j}(\alpha_{k,ij}) - \beta_{k,ij}^G) \mathbf{G}_{k,ij} \\ & + \sum_{\Omega_{BF}} \pm \alpha_{k,ij} \mathbf{F}_{k,ij} + \sum_{\Omega_{BG}} \pm \alpha_{k,ij} \mathbf{G}_{k,ij}. \end{aligned}$$

Here $\beta_{k,ij}^F$ and $\beta_{k,ij}^G$ are the sum of all terms which contribute to $\mathbf{F}_{k,ij}$ and $\mathbf{G}_{k,ij}$, respectively,

$$(19) \quad \beta_{k,ij}^F = \sum_{mn} c_{k',mn,k,ij}^{FF} \alpha_{k',mn} \gamma_{k',mn,k,ij}^F + c_{k',mn,k,ij}^{FG} \alpha_{k',mn} \gamma_{k',mn,k,ij}^G,$$

$$(20) \quad \beta_{k,ij}^G = \sum_{mn} c_{k',mn,k,ij}^{GF} \alpha_{k',mn} \gamma_{k',mn,k,ij}^F + c_{k',mn,k,ij}^{GG} \alpha_{k',mn} \gamma_{k',mn,k,ij}^G,$$

where

$$\begin{aligned} c_{k',mn,k,ij}^{FF} &= \pm K \{ +\partial y_k / \partial S \partial x_{k'} / \partial R - \partial x_k / \partial S \partial y_{k'} / \partial R \} \frac{1}{J_{k'}}, \\ c_{k',mn,k,ij}^{FG} &= \pm K \{ +\partial y_k / \partial S \partial x_{k'} / \partial S - \partial x_k / \partial S \partial y_{k'} / \partial S \} \frac{1}{J_{k'}}, \\ c_{k',mn,k,ij}^{GF} &= \pm K \{ -\partial y_k / \partial R \partial x_{k'} / \partial R + \partial x_k / \partial R \partial y_{k'} / \partial R \} \frac{1}{J_{k'}}, \\ c_{k',mn,k,ij}^{GG} &= \pm K \{ -\partial y_k / \partial R \partial x_{k'} / \partial S + \partial x_k / \partial R \partial y_{k'} / \partial S \} \frac{1}{J_{k'}}, \\ K &= \frac{1}{2} \quad \text{or } 1. \end{aligned}$$

The constant K is equal to 1 except at a boundary where it will be equal to $\frac{1}{2}$ for the following reason: The flux that is interpolated at the vertex is averaged to the adjoining faces by (17) and (18). There will normally be two faces adjoining each vertex that is interpolated, in which case the flux at the vertex is used twice and $K = 1$. When the interpolation boundary intersects a true boundary, however, the vertex that lies on the true boundary will only have one face adjacent to it, in which case $K = \frac{1}{2}$.

We require that dS/dt be equal to some approximation to the integral of the fluxes through the boundary of Ω . Thus we set to zero the coefficients of the interior fluxes. Thus the constraints on the $\alpha_{k,ij}$ are of the following types

1. For those α belonging to interior cells

$$(21) \quad \Delta_{-i}(\alpha_{k,ij}) = 0 \quad \text{or}$$

$$(22) \quad \Delta_{-j}(\alpha_{k,ij}) = 0 \quad \text{for } k, ij \in \Omega_I.$$

2. For those cells that are interpolated from

$$(23) \quad \Delta_{-i}(\alpha_{k,ij}) = \beta_{k,ij}^F \quad \text{for } k, ij \in \Omega_F,$$

$$(24) \quad \Delta_{-j}(\alpha_{k,ij}) = \beta_{k,ij}^G \quad \text{for } k, ij \in \Omega_G.$$

We will call (21)–(24) the *conservation constraints*. These equations impose conditions on the integration weights $\alpha_{k,ij}$ and the interpolation coefficients (that appear implicitly through

β^F and β^G). As long as these equations are satisfied we will have a conservative scheme. We are free to add extra equations to specify the interpolation coefficients, as long as these new equations are consistent with the conservation constraints.

3.2. Interpolee cells and the conservation constraints. In Fig. 10(a) we show an overlapping grid consisting of a rotated square (grid $k = 2$) sitting on an unrotated square (grid $k = 1$). The edge of the rotated square is interpolated from points on the unrotated square.

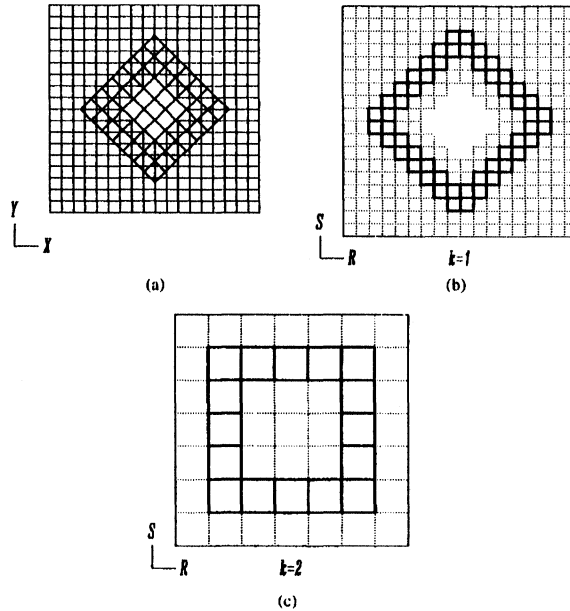


FIG. 10. Overlapping grid and interpolator cells on each component grid.

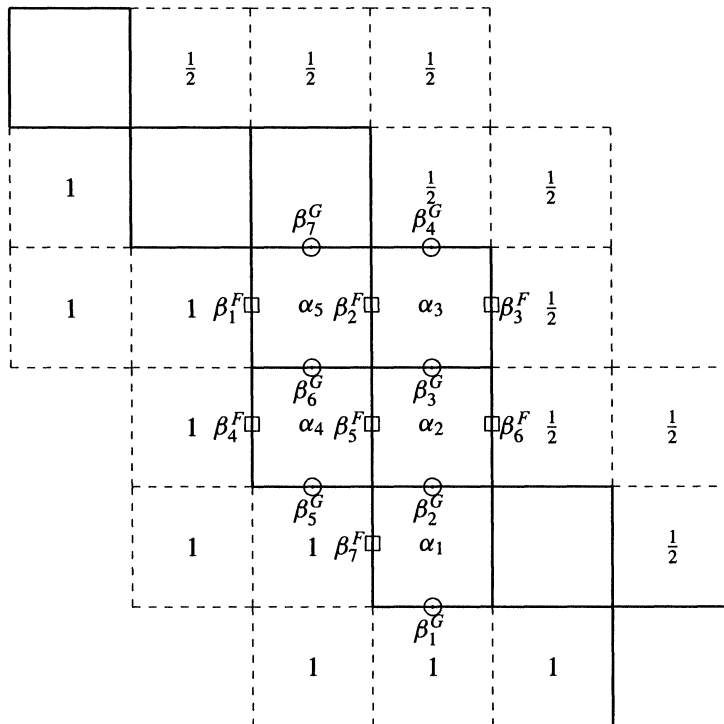
The unrotated square has points near the centre that are not used. The points on the boundary of this cut-out portion of grid 1 are interpolated from the rotated square. Since we often refer to those points or cells that are used in the interpolation of another point, we define such a point or cell to be an interpolator point or interpolator cell.

DEFINITION 3.1. A point on grid k that is used to obtain an interpolated value on grid k' is called an interpolator point of grid k . Cells that are completely surrounded by interpolator points are called interpolator cells.

In Figs. 10(b) and 10(c), we show the location of the interpolator cells for component grid 1 and component grid 2 in the (R, S) unit squares. The interpolator cells are outlined with thick lines. Interpolator cells are those cells that are used by another grid for interpolation. Thus the values on the vertices of the interpolator cells on grid 1 are used to interpolate a value to the interpolation points of grid 2. Note that the set of interpolator cells partitions the grid into distinct regions. This partitioning property is a necessary condition for the existence of a conservative interpolation scheme. A cell that is not an interpolator cell will have a value of $\alpha = 1$ if it is not in an overlapping region and a value of $\alpha = \frac{1}{2}$ if it is in a region where two grids overlap. Thus the dotted cells exterior to the region bounded by the interpolator cells on grid 1 will have $\alpha = 1$ while the dotted cells inside this region will have $\alpha = \frac{1}{2}$. Conversely, the dotted cells interior to the region bounded by interpolator cells on grid 2 will have $\alpha = 1$ while outside this region $\alpha = \frac{1}{2}$. An interpolator cell may have a value for α that is different from 1 or $\frac{1}{2}$. This

value is determined in the process of determining the conservative interpolation coefficients.

In Fig. 11 we show, in detail, a set of interpolatee cells. For some of these cells we have marked the unknown α 's and β 's that must be determined. It is assumed that the region to the upper right of the figure lies in an overlapping region and thus $\alpha = \frac{1}{2}$. The region to the lower left is a non-overlapping region and $\alpha = 1$. Recall that each interpolatee cell must satisfy the conditions given by the conservation constraints, (21)–(24). At the bottom of the figure we list some of these equations that must be satisfied in order that the interpolation be conservative.



$$\begin{aligned} \alpha_4 - 1 &= \beta_5^G & \alpha_2 - \alpha_1 &= \beta_2^G & \alpha_3 - \alpha_2 &= \beta_3^G, \\ \alpha_4 - 1 &= \beta_4^F & \alpha_2 - \alpha_4 &= \beta_5^F & \alpha_3 - \alpha_5 &= \beta_2^F, \\ \frac{1}{2} - \alpha_2 &= \beta_6^F & \frac{1}{2} - \alpha_3 &= \beta_3^F, \\ \frac{1}{2} - \alpha_3 &= \beta_4^G. \end{aligned}$$

FIG. 11. Some interpolatee cells and some sample conservation constraints.

3.3. Accuracy and smoothness of the interpolation coefficients. The enforcement of conservation constraints impose certain restrictions on the interpolation coefficients. To fully specify the interpolation coefficients it is necessary to add further constraints. These constraints can be chosen for accuracy requirements. We may also want to impose smoothness requirements on the interpolation coefficients since coefficients that have large negative and positive values do not enhance the stability characteristics of a scheme.

Accuracy. It is natural to impose conditions to increase the accuracy of the interpolation. Suppose we want to interpolate a function $f(r, s)$ at some point (r_c, s_c) in terms of the values of f at a set of points $\{f(r_{mn}, s_{mn})\}$:

$$f(r_c, s_c) \approx \sum_{mn} \gamma_{mn} f(r_{mn}, s_{mn}).$$

By Taylor's formula

$$\begin{aligned} f(r_{mn}, s_{mn}) &= f(r_c, s_c) + (r_{mn} - r_c) f_r(r_c, s_c) + (s_{mn} - s_c) f_s(r_c, s_c) \\ &\quad + (r_{mn} - r_c)^2 \frac{1}{2} f_{rr}(r_c, s_c) + (r_{mn} - r_c)(s_{mn} - s_c) f_{rs}(r_c, s_c) + \cdots \end{aligned}$$

and thus

$$\begin{aligned} f(r_c, s_c) &= \left\{ \sum_{mn} \gamma_{mn} \right\} f(r_c, s_c) + \left\{ \sum_{mn} \gamma_{mn} (r_{mn} - r_c) \right\} f_r(r_c, s_c) \\ &\quad + \left\{ \sum_{mn} \gamma_{mn} (s_{mn} - s_c) \right\} f_s(r_c, s_c) + \left\{ \sum_{mn} \gamma_{mn} (r_{mn} - r_c)^2 \right\} \frac{1}{2} f_{rr}(r_c, s_c) + \cdots. \end{aligned}$$

Thus to interpolate to an accuracy of $O(h^P)$ we require that the interpolation coefficients satisfy

$$\begin{aligned} \sum_{mn} \gamma_{mn} &= 1 + O(h^P), \\ \sum_{mn} \gamma_{mn} (r_{mn} - r_c)^p (s_{mn} - s_c)^q &= O(h^P) \quad \text{for } p + q < P. \end{aligned}$$

Smoothness. There are a number of ways to impose smoothness constraints on the interpolation coefficients. Here is one reasonable way: if we think of representing the interpolation coefficients as a Fourier sine series then we can choose to make the coefficients of the highest frequencies in this series vanish. In particular, consider the coefficients $\gamma_{k',mn,k,ij}^F$ for a given interpolation point (i.e., for fixed (k', k, ij)). Then we write

$$\gamma_{mn}^F = \sum_{i=1}^M \sum_{j=1}^N \hat{\gamma}_{ij}^F \sin\left(\frac{i m \pi}{M+1}\right) \sin\left(\frac{j n \pi}{N+1}\right),$$

where for clarity we drop some of the subscripts. The Fourier coefficients $\hat{\gamma}_{ij}^F$ are given by

$$\hat{\gamma}_{ij}^F = \frac{4}{M N} \sum_{m=1}^M \sum_{n=1}^N \gamma_{mn}^F \sin\left(\frac{i m \pi}{M+1}\right) \sin\left(\frac{j n \pi}{N+1}\right).$$

The first smoothness constraint is obtained by setting $\hat{\gamma}_{MN}^F = 0$, followed by $\hat{\gamma}_{M-1N}^F = 0$, $\hat{\gamma}_{MN-1}^F = 0$, and so on.

3.4. Determination of the conservative interpolation coefficients. Here is an outline of the algorithm we use to compute the conservative interpolation coefficients. We work on one component grid at a time. We determine the

$$\alpha_{kij}, \beta_{kij}^F, \beta_{kij}^G, \gamma_{k',mn,k,ij}^F \quad \text{and} \quad \gamma_{k',mn,k,ij}^G$$

for a given component grid k . Note that we determine the interpolation coefficients for points which interpolate from grid k as opposed to the interpolation points on grid k .

Step 1. Mark the interpolatee cells. Mark the cells used for interpolation. Make sure that these interpolatee cells divide the grid into separated regions. The algorithm for marking the interpolatee cells follows.

1. For each interpolation point on grid $k' \neq k$, mark the cells on grid k which are used for interpolation.

2. For neighbouring interpolation points, check whether the set of interpolatee cells used by one interpolation point is connected to the set of interpolatee cells used by the second interpolation point.

3. If the sets of interpolatee cells are not connected, add extra interpolation coefficients to each of the two interpolation formulae so that the interpolatee cells used by the added coefficients will connect the two sets of interpolatee cells.

Step 2. Form the β equations. Determine the equations for $\beta_{kij}^{F,G}$ using (19) and (20). This defines the β 's in terms of the interpolation coefficients γ :

$$\begin{aligned}\beta^F &= B^F(\gamma^F, \gamma^G), \\ \beta^G &= B^G(\gamma^F, \gamma^G),\end{aligned}$$

or in matrix form,

$$\begin{bmatrix} I & 0 & A_{13} \end{bmatrix} \begin{bmatrix} \beta \\ \alpha \\ \gamma \\ \vdots \\ \gamma \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix}.$$

Step 3. Conservation constraints. Form the conservation constraints (23)–(24) for interpolatee points

$$\begin{aligned}\Delta_{-i}(\alpha_{k,ij}) &= \beta_{k,ij}^F \quad \text{or} \\ \Delta_{-j}(\alpha_{k,ij}) &= \beta_{k,ij}^G \quad \text{for interpolatee cells on grid } k\end{aligned}$$

with boundary conditions

$$\begin{aligned}\alpha_{k,ij} &= 1 \quad \text{in nonoverlapping regions,} \\ \alpha_{k,ij} &= \frac{1}{2} \quad \text{where two grids overlap.}\end{aligned}$$

By taking $\alpha_{k,ij} = \frac{1}{2}$ where two component grids overlap we choose to equally weight cells of the same area on any two grids. This corresponds to choosing $\mu = \nu = \frac{1}{2}$ in the one-dimensional case. If three component grids all overlap in a common region then the boundary condition would be taken as $\alpha_{k,ij} = \frac{1}{3}$. At this point we have a system of equations, which has the form

$$\begin{bmatrix} I & 0 & A_{13} \\ A_{21} & A_{22} & 0 \end{bmatrix} \begin{bmatrix} \beta \\ \alpha \\ \gamma \\ \vdots \\ \gamma \end{bmatrix} = \begin{bmatrix} 0 \\ a \end{bmatrix}.$$

Step 4. Initial factor. Factor the matrix to eliminate A_{21} and to put A_{22} into upper triangular form:

$$\begin{bmatrix} I & 0 & A_{13} \\ 0 & U_{22} & A_{23} \\ 0 & 0 & A_{33} \end{bmatrix} \begin{bmatrix} \beta \\ \alpha \\ \gamma \\ \vdots \\ \gamma \end{bmatrix} = \begin{bmatrix} 0 \\ a \\ b \end{bmatrix}.$$

After this step the β 's have been eliminated. U_{22} is a square upper triangular matrix that defines the area weights α in terms of the γ 's. The rectangular matrix A_{33} multiplies the γ 's and represents the constraints that must be satisfied by the γ 's so that the interpolation is conservative. At this point we need only to consider the constraint equations for the γ 's:

$$(25) \quad A_{33}x = b : \quad \begin{bmatrix} A_{33} \end{bmatrix} \begin{bmatrix} \gamma \\ \vdots \\ \gamma \end{bmatrix} = \begin{bmatrix} b \end{bmatrix}.$$

The problem now is to choose accurate interpolation coefficients that are consistent with the above conservation constraints.

Step 5. Interpolation accuracy and smoothness. For each point (k', i, j) interpolating from grid k there is a set of interpolation coefficients $\gamma_{k',ij,mn}^F$ and $\gamma_{k',ij,mn}^F$ defining the interpolation of the fluxes. For each interpolation formula we construct sufficiently many equations based on accuracy and smoothness to determine the interpolation coefficients completely (ignoring the fact that the coefficients will be constrained by conservation). Our choice of equations is described in §3.3. If we write the equations that determine all γ 's in one large matrix equation it will have the form of a block diagonal system:

$$(26) \quad Gx = r : \quad \begin{bmatrix} G_1 & 0 & 0 & \dots & 0 \\ 0 & G_2 & 0 & \dots & 0 \\ 0 & 0 & G_3 & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 \\ 0 & 0 & \dots & 0 & G_N \end{bmatrix} \begin{bmatrix} \gamma_1 \\ \gamma_2 \\ \gamma_3 \\ \vdots \\ \gamma_N \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \\ r_3 \\ \vdots \\ r_N \end{bmatrix}.$$

Each diagonal block G_i is square and invertible. A typical dimension for a block G_i is 6×6 , corresponding to the interpolation shown in Fig. 9.

Step 6. Solution method. It is now necessary to find a *good* solution to the over-determined system consisting of the conservation constraints, (25), and (26) defining the interpolation coefficients in terms of accuracy and smoothness. There are various possible approaches to this problem; we present two here.

Solution method I. In method I we augment the constraint equations $A_{33}x = b$ by sufficiently many accuracy and smoothness equations from the matrix G so that the resulting system is nonsingular. This system can be solved to determine the interpolation coefficients. This method is harder to implement in practice than it sounds. The equations taken from G may be dependent on those found in A_{33} or even slightly inconsistent with those equations. Thus it is not easy to know which set of equations should be used. We have been unsuccessful at devising a stable and efficient technique for this approach and thus we have devised method II.

Solution method II. In method II we solve the constrained least squares problem

$$(27) \quad \min_{x \in \mathbb{R}^n} \|Gx - r\| \quad \text{subject to } A_{33}x = b.$$

To be specific, assume that $A_{33} \in \mathbb{R}^{m \times n}$ and $G \in \mathbb{R}^{n \times n}$. The number of constraints m is assumed to be less than the number of interpolation coefficients n . This method generates a solution that is exactly conservative but only satisfies the accuracy and smoothness equations in a least squares sense. However, by weighting the accuracy equations more than the smoothness equations it is possible, in practice, to obtain accurate interpolation equations with no large interpolation coefficients. If the interpolation is intended to be accurate to $O(h^p)$ then we scale the smoothness constraints so that the l_2 norms of the rows corresponding to the smoothness equations are $O(h^p)$. The accuracy equations, as defined by (25), are naturally scaled in the proper way. Note that if we were only interested in weighting all the equations in $Gx = r$ the same then we could replace $\min_{x \in \mathbb{R}^n} \|Gx - r\|$ in (27) with $\min_{x \in \mathbb{R}^n} \|x - G^{-1}r\|$. The solution to this new problem is somewhat simpler than the solution to (27) that is outlined below.

We can solve problem (27) as follows. Since the matrix G is nonsingular we can make the change of variables $x = G^{-1}y$ to give an equivalent formulation of the problem:

$$\min_{y \in \mathbb{R}^n} \|y - r\| \quad \text{subject to } A_{33}G^{-1}y = b.$$

Recall that G is block diagonal and thus its inverse is quickly computed. Letting $A = A_{33}G^{-1} \in \mathbb{R}^{m \times n}$ be the new constraint matrix, then our problem becomes

$$(28) \quad \min_{y \in \mathbb{R}^n} \|y - r\| \quad \text{subject to } Ay = b.$$

The transpose of A can be factored using the QR decomposition method to give

$$A^T = Q \begin{bmatrix} R \\ 0 \end{bmatrix},$$

where $R \in \mathbb{R}^{m \times m}$ is an upper triangular matrix, which is nonsingular if A has full rank. If we partition the orthogonal matrix $Q \in \mathbb{R}^{n \times n}$ into $Q = [Q_1 \ Q_2]$, with $Q_1 \in \mathbb{R}^{n \times m}$ consisting of the first m columns, then the general solution to the underdetermined system $Ay = b$ is

$$y = Q_1(R^T)^{-1}b + Q_2w,$$

where $w \in \mathbb{R}^{n-m}$ is arbitrary. Thus the solution of the constrained least squares problem (28) is found when w is the solution of the following least squares problem:

$$\min_{w \in \mathbb{R}^{n-m}} \|Q_2w - (r - Q_1(R^T)^{-1}b)\|.$$

The solution to this problem is

$$w_{ls} = Q_2^T(r - Q_1(R^T)^{-1}b) = Q_2^Tr,$$

since $Q_2^TQ_2 = I$ and $Q_2^TQ_1 = 0$. Therefore the constrained least squares solution y_{cls} is

$$\begin{aligned} y_{cls} &= Q_1(R^T)^{-1}b + Q_2w_{ls} \\ &= [Q_1 \ Q_2] \begin{bmatrix} (R^T)^{-1}b \\ Q_2^Tr \end{bmatrix} = Q \begin{bmatrix} (R^T)^{-1}b \\ Q_2^Tr \end{bmatrix}, \end{aligned}$$

and the solution to (27) is

$$x_{\text{cls}} = G^{-1}y_{\text{cls}} = G^{-1}Q \begin{bmatrix} (R^T)^{-1}b \\ Q_2^T r \end{bmatrix}.$$

Although the matrix $Q \in \mathbb{R}^{n \times n}$ is large, it can be stored in an efficient manner as a product of m Householder matrices. See, for example, the discussion in the LINPACK Users' Guide [9]. The cost to compute x_{cls} is $O(m^2n)$ floating-point operations and $O(mn)$ words of storage.

4. Numerical experiments with conservative interpolation. In this section we describe a method for discretizing conservation laws in a conservative manner on overlapping grids, and we present numerical results that show the propagation of a shock through an overlapping grid. We study a two-dimensional analogue of Burger's equation,

$$(29) \quad \frac{\partial u}{\partial t} + \frac{\partial}{\partial x} \left(\frac{1}{2} u^2 \right) = \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$$

with the initial condition

$$u(x, y, 0) = u_0(x) := c - \tanh \frac{x - x_0}{2\nu}.$$

The exact solution to this problem is a shock layer moving to the right with speed c ,

$$u(x, y, t) = u_0(x - ct).$$

To compute a numerical solution to this problem on the rectangle $-\frac{3}{2} \leq x \leq \frac{3}{2}$, $-1 \leq y \leq 1$, we choose $x_0 = -1$ so that the shock is initially within the domain under consideration. We use periodic boundary conditions at $y = \pm 1$ and Dirichlet boundary conditions

$$u \left(\pm \frac{3}{2}, y, t \right) = u_0 \left(\pm \frac{3}{2} - ct \right)$$

at $x = \pm \frac{3}{2}$. In particular, we consider the case $c = \frac{1}{50}$ and compute the solution for $0 \leq t \leq 100$, so that the shock should move from $x = -1$ at $t = 0$ to $x = 1$ at $t = 100$. We discretize the problem on overlapping grids consisting of an overall uniform rectangular grid covering the entire domain and a smaller "obstacle" grid located approximately in the centre of the domain. The shock starts to the left of the obstacle at $t = 0$ and should have passed entirely through it when $t = 100$.

4.1. Discretization on a curvilinear grid. The two-dimensional Burger's equation (29) is of the form of a system of conservation laws

$$\frac{\partial u}{\partial t} + \frac{\partial}{\partial x} f(u) + \frac{\partial}{\partial y} g(u) = 0,$$

where the flux functions are given by

$$f(u) = f_c(u) - \nu \frac{\partial u}{\partial x}, \quad f_c(u) = \frac{1}{2} u^2, \quad g(u) = g_c(u) - \nu \frac{\partial u}{\partial y}, \quad g_c(u) = 0.$$

Before discretizing this equation on a curvilinear grid, we transform it into a coordinate system where the grid is a uniform rectangular grid,

$$\frac{\partial u}{\partial t} + \frac{1}{J} \left(\frac{\partial}{\partial r} F(u) + \frac{\partial}{\partial s} G(u) \right) = 0,$$

where

$$J = x_r y_s - y_r x_s,$$

$$F(u) = y_s f(u) - x_s g(u) = y_s f_c(u) - x_s g_c(u) + \frac{\nu}{J} \left((x_r x_s + y_r y_s) \frac{\partial u}{\partial s} - (x_s^2 + y_s^2) \frac{\partial u}{\partial r} \right),$$

$$G(u) = x_r g(u) - y_r f(u) = x_r g_c(u) - y_r f_c(u) + \frac{\nu}{J} \left((x_r x_s + y_r y_s) \frac{\partial u}{\partial r} - (x_r^2 + y_r^2) \frac{\partial u}{\partial s} \right).$$

The overlapping grid-generation program CMPGRD represents a curvilinear grid as a uniform grid on the unit square and a smooth transformation $\mathbf{d}(r, s) : [0, 1]^2 \rightarrow \mathbb{R}^2$, so that the gridpoints of the curvilinear grid are $(x_{ij}, y_{ij}) = \mathbf{d}(i\Delta r, j\Delta s)$, where $\Delta r = 1/(N_r - 1)$ and $\Delta s = 1/(N_s - 1)$. CMPGRD also provides the derivative

$$\begin{bmatrix} x_r & x_s \\ y_r & y_s \end{bmatrix}_{ij} = \frac{\partial \mathbf{d}}{\partial(r, s)}(i\Delta r, j\Delta s)$$

at the gridpoints (cell vertices). The subscript ij is assumed for x_r, x_s, y_r , and y_s wherever they appear below. To aid in the discretization, we define the averaging operators

$$\mu_{-i} u_{ij} = \frac{1}{2}(u_{i-1,j} + u_{ij}), \quad \mu_{+i} u_{ij} = \frac{1}{2}(u_{ij} + u_{i+1,j}),$$

$$\mu_{-j} u_{ij} = \frac{1}{2}(u_{i,j-1} + u_{ij}), \quad \mu_{+j} u_{ij} = \frac{1}{2}(u_{ij} + u_{i,j+1}),$$

the undivided difference operators

$$\Delta_{-i} u_{ij} = u_{i,j} - u_{i-1,j}, \quad \Delta_{+i} u_{ij} = u_{i+1,j} - u_{i,j}, \quad \Delta_{0i} u_{ij} = \frac{1}{2}(u_{i+1,j} - u_{i-1,j}),$$

$$\Delta_{-j} u_{ij} = u_{i,j} - u_{i,j-1}, \quad \Delta_{+j} u_{ij} = u_{i,j+1} - u_{i,j}, \quad \Delta_{0j} u_{ij} = \frac{1}{2}(u_{i,j+1} - u_{i,j-1}),$$

and the cell-centred quantities

$$A_{ij} = \mu_{+i} \mu_{+j} \left(\frac{1}{\Delta r \Delta s J_{ij}} \right), \quad B_{ij} = \mu_{+i} \mu_{+j} \left(\frac{x_r x_s + y_r y_s}{J_{ij}} \right).$$

A_{ij} is an approximation to second-order accuracy of the reciprocal of the area of the cell (i, j) . Then we discretize equation (29) in space to second-order accuracy as

$$(30) \quad \frac{du_{ij}}{dt} + A_{ij}(\Delta_{+i} F_{ij} + \Delta_{+j} G_{ij}) = 0,$$

where

$$(31) \quad F_{ij} = \Delta s [(\mu_{+j} y_s)(\mu_{-i} f_c(u_{ij})) - (\mu_{+j} x_s)(\mu_{-i} g_c(u_{ij}))] \\ + \nu \{ (\mu_{-i} B_{ij})(\mu_{-i} \Delta_{0j} u_{ij}) - \Delta s^2 (\mu_{-i} A_{ij}) [(\mu_{+j} x_s)^2 + (\mu_{+j} y_s)^2] (\Delta_{-i} u_{ij}) \},$$

$$(32) \quad G_{ij} = \Delta r [(\mu_{+i} x_r)(\mu_{-j} g_c(u_{ij})) - (\mu_{+i} y_r)(\mu_{-j} f_c(u_{ij}))] \\ + \nu \{ (\mu_{-j} B_{ij})(\mu_{-j} \Delta_{0i} u_{ij}) - \Delta r^2 (\mu_{-j} A_{ij}) [(\mu_{+i} x_r)^2 + (\mu_{+i} y_r)^2] (\Delta_{-j} u_{ij}) \}.$$

4.2. Discretization on an overlapping grid. To discretize (29) on an overlapping grid, we must supply interpolation boundary conditions in the regions of overlap between grids. The program CMPGRD is designed to generate overlapping grids where the solution u_{ij} is defined either at cell-centres or at cell vertices. In either case it guarantees that all eight neighbours of any “discretization point” are either discretization points themselves or they are “interpolation points” where function values may be interpolated from the corresponding function values at discretization points of another grid. We have implemented conservative interpolation of fluxes using “cell-vertex overlapping grids,” as explained in §3. On such a composite grid, CMPGRD labels the vertices as either discretization points, interpolation points, or unused points. We discretize the problem with u_{ij} represented at cell-centres and the fluxes represented on the sides of each cell, as shown in Fig. 12. The computation of F_{ij}

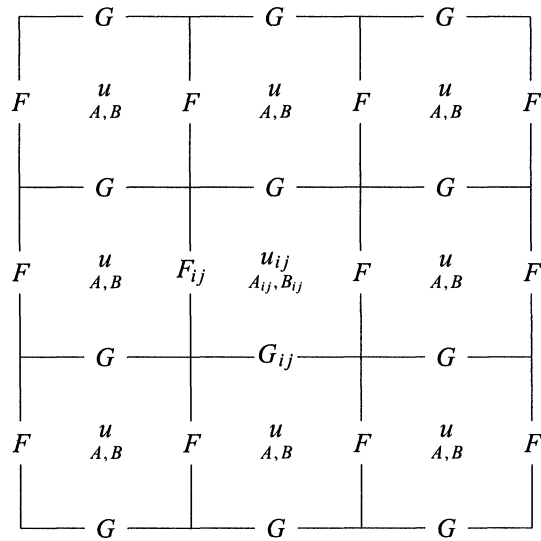


FIG. 12. Grid functions.

uses the stencil consisting of $u_{i-1,j}$, $u_{i-1,j\pm 1}$, u_{ij} and $u_{i,j\pm 1}$. Suppose u is known at every cell-centre that is surrounded by vertices all of which are labelled as either discretization points or interpolation points. CMPGRD guarantees that all eight neighbours of a discretization point are either discretization points or interpolation points. Therefore we can compute F_{ij} on any cell side that connects two discretization points (i, j) and $(i, j + 1)$. Similarly, we can compute G_{ij} on any cell side that connects two discretization points (i, j) and $(i + 1, j)$. We can update u_{ij} using (30) anywhere that we can compute the expression for du_{ij}/dt ; for this, we need values for F_{ij} , $F_{i+1,j}$, G_{ij} , and $G_{i,j+1}$. We know how to compute these fluxes for any cell surrounded by discretization points. To update u in a cell, one or more of whose neighbouring vertices is an interpolation point, we must first interpolate the fluxes that we could not otherwise compute. We must interpolate the flux on each cell side adjoining two vertices, at least one of which is labelled as an interpolation point and the other is labelled as either an interpolation point or a discretization point. We interpolate these fluxes by the procedure explained in §3.1.

We use the classical fourth-order accurate four-stage Runge–Kutta timestepping method. We choose the timestep to be close to the stability limit, which we determined experimentally. In particular, we use

$$\Delta t = \min\{.8h, .3\frac{h^2}{\nu}\}, \quad \text{where } h = \min\{\sqrt{x_r^2 + y_r^2}\Delta r, \sqrt{x_s^2 + y_s^2}\Delta s\}.$$

The minimum here is taken over all gridpoints on all component grids. In the numerical examples we took $\nu = .01$, which resulted in sharp shocks with little or no overshoot.

4.3. Numerical results. Figure 13 shows the numerical solution at time $t = 0, 20, \dots, 100$ for the case $c = \frac{1}{50}$ of a slowly-moving shock, obtained using an overlapping grid

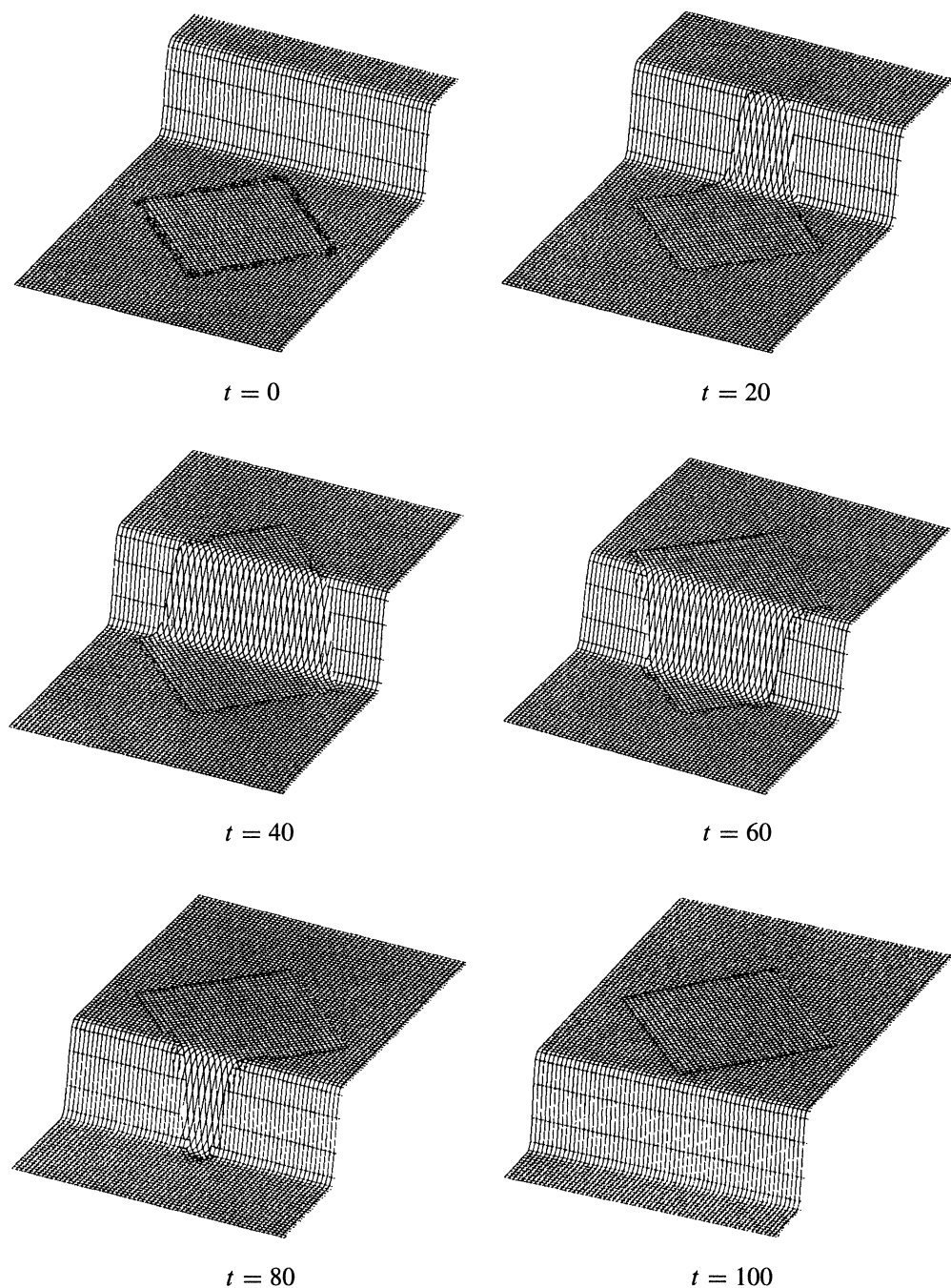


FIG. 13. Numerical solution at times $t = 0, 20, \dots, 100$.

consisting of an overall rectangular grid of 90 by 60 square cells and a square grid in the centre of the region, with sides of length 1.2, rotated 45° , with 36×36 cells of the same size as those of the overall grid. The numerical solution shows an undistorted shock whose location is correct to within the width of one grid cell at all times. Figure 14 is a graph of the

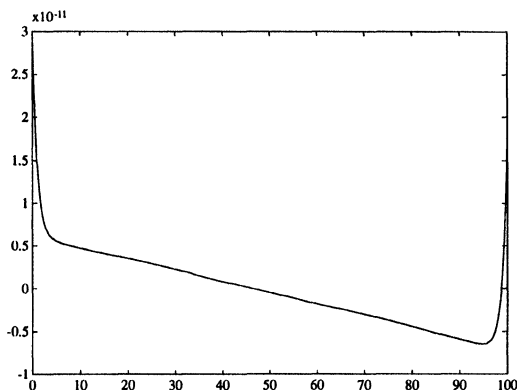


FIG. 14. Conservation error as a function of time.

difference between the conserved discrete integral of u and the analytic integral of the exact solution. The mean value of this conservation error, 3.26×10^{-6} , has been subtracted out, as this error is only due to truncation error in the discrete integral of the initial condition. The resulting conservation error is less than 3×10^{-11} at all times. It changes most rapidly at $t = 0$ and $t = 100$, when the shock is nearest to the domain boundaries at $x = \pm \frac{3}{2}$. This effect is due to the truncation error in the discretization of the boundary conditions and not to the interpolation of fluxes between grids.

5. Conclusions. We have described a scheme for interpolating in a conservative manner when solving systems of conservation laws on overlapping grids. The technique is described in one and two space dimensions; the extension to three dimensions is straightforward. We consider the interpolation coefficients and the integration weights in the conserved integral as free parameters. The condition that the scheme be conservative imposes constraints on these parameters. We have shown how to derive these constraints and how to determine accurate interpolation coefficients consistent with the conservation constraints. This requires the solution of a system of equations with number of unknowns essentially equal to the number of interpolation coefficients on one component grid. We have written a code to generate the conservative interpolation coefficients for two-dimensional grids generated by the overlapping grid generation program, CMPGRD.

We have numerically solved a two-dimensional viscous Burger's equation using conservative interpolation and have demonstrated the exact conservative nature of the method. In future work we hope to solve the full Navier–Stokes or Euler equations using the conservative interpolation and to study the problem of slowly moving shocks passing through the interface between grids.

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